National Institute of Open Schooling
Senior Secondary Course: Mathematics
Lesson 10: Principle of Mathematical Induction
Worksheet -10

1 Write any five statements in Mathematics.

2 Differentiate between direct proof and proof by Mathematical induction in Mathematics with examples.

3 If P (n) is a statement $2^{3n}-1$ is an integral multiple of 7, verify that P (1) and P (2) and P (3) are true.

4 Prove that $1+2+3+.....+n=\frac{n(n+1)}{2}$, $n \in N$

5 By using the principle of mathematical induction:

$1+3+3^2+.....+3^{n-1}=\frac{3^n-1}{2}$, For all $n \in N$.

6 Find all natural numbers n , prove that $3^n > n$.

7 Prove that $n(n+1)(n+5)$ is a multiple of 3, for all $n \in N$ by principle of mathematical induction.

8 Prove that $\frac{1}{1\times 2}+\frac{1}{2\times 3}+.....+\frac{1}{n(n+1)}=\frac{n}{n+1}$

For $n \in N$

9 For all natural numbers n ∈ N, prove that n < 2^n by mathematical induction.

10 By principle of mathematical induction for all $n \in N$:
$n(n+1)(n+2)$ is divisible by 6.